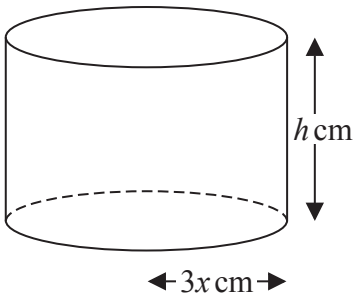
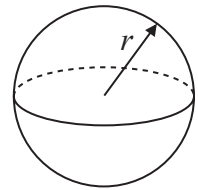
**17** The diagram shows a solid metal cylinder.




Volume of sphere = $\frac{4}{3}\pi r^3$




The cylinder has base radius $3x$ cm and height $h$ cm.

The metal cylinder is melted.
All the metal is then used to make 270 spheres.

Each sphere has a radius of $\frac{1}{2}x$ cm.

Find an expression, in its simplest form, for $h$ in terms of $x$.

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**(Total for Question 17 is 3 marks)**